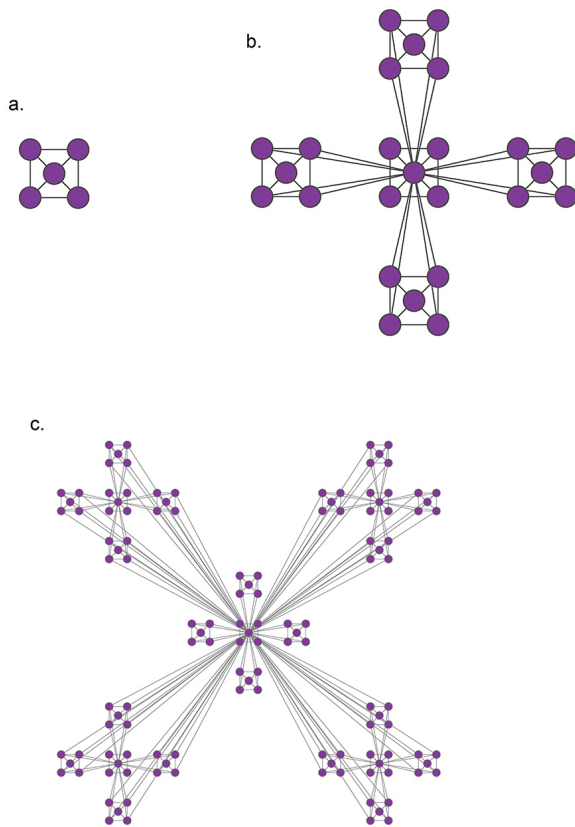


L7: Communities Within Communities



A hierarchical model that generates a scale-free network with degree exponent. (Image 9.13 from networksciencebook.com)

density of the connections at that level is lower than the density at the previous level.

This is of course an extreme example of hierarchical modularity – the connections are deterministic and the connectivity pattern is the same at every layer. In practice, most networks have some degree of randomness and the depth of these hierarchies may not exceed a certain maximum level for practical reasons (*for example: what is the maximum hierarchical depth in your professional organization or university?*)

Nevertheless, the deterministic structure of this toy network allows us to derive mathematically an interesting property of this network: namely, the clustering coefficient of a node in this network drops proportionally to the inverse of the node's degree:

$$C(k) \approx \frac{2}{k}$$

We will not show the derivation here – it is simple however and you can find it in your textbook.

This is an interesting “**signature**” for this hierarchically modular network – the more connected a node is, the less interconnected its neighbors are. This is very different than random networks, without any community structure, in which the clustering coefficient is independent of the degree.

In many real-world networks, we observe communities within communities within communities, etc. For example, in a professional organization, we may observe small communities of 3-4 people that work together on a specific component of a large project, nested within a larger community of 20-30 members working on that the project, nested within an entire department of 100+ people, etc.

This is referred to as **Hierarchical Modularity**. Each module corresponds to a community of nodes. Smaller, more tightly connected communities can be members of larger but less tightly connected larger communities. This recursive process can continue for multiple levels until we reach a level in which the group of nodes is so loosely connected that they do not form a community anymore.

A prototypical example of such a hierarchically modular architecture is illustrated in this visualization. The most elementary community is a five-node clique module (*part a*). *Note that the diagonal elements are also connected even though it is not very clear in the illustration.*

At the next level (*part b*), we see a similar structure that is composed of five of those cliques, with the clique at the middle playing the role of the “connector” for the four peripheral cliques. Note that the network at this level is NOT a clique, and so it is less tightly connected than the five smaller communities it is composed of.

At the third level of the hierarchy (*part c*), we connect in a similar manner five instances of the part-b module. Again, the module at the center is the connector of the four peripheral modules, and again the density of the connections at that level is lower than the density at the previous level.

Food For Thought

Please go through the derivations for the above clustering coefficient in your textbook.